Hints: Functional and differential equations – 2. Functional-differential equations

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Put $a = t+h$, $b = t-h$: $f(t+h) - f(t-h) = 2hf'(t)$. With $h = 1$, $f'(t) = \frac{f(t+1)-f(t-1)}{2}$ is continuous, and in the same way differentiable: by induction $f \in C^\infty$.

Hint 2. Differentiate the identity twice in h: $f''(t+h) - f''(t-h) = 0$ for all t, h, so f'' is constant. Answer: $f(x) = ax^2 + bx + c$.

Problem 2

Hint 1. Regularity first: the geometric mean of $2x$ and $\frac{x}{2}$ is x, so $f'(x) = \frac{f(2x)-f(x/2)}{3x/2}$ and $f \in C^\infty$. Put $a = ts$, $b = \frac{t}{s}$: $f(ts) - f(\frac{t}{s}) = t(s - \frac{1}{s})f'(t)$.

Hint 2. Differentiate in s: with $\psi(x) = xf'(x)$ you get $\psi(ts) + \psi(\frac{t}{s}) = (s + \frac{1}{s})\psi(t)$. Differentiate twice more in s and put $s = 1$: $t^2\psi'' + t\psi' - \psi = 0$.

Hint 3. This Euler equation gives $\psi(t) = At + \frac{B}{t}$, i.e. $f'(x) = A + \frac{B}{x^2}$. Answer: $f(x) = Ax + \frac{C}{x} + D$; check: $f(a) - f(b) = (a-b)(A - \frac{C}{ab})$.

Problem 3

Hint 1. Put $c = f(0)$, $d = f'(0)$; $y = 0$ gives $(1-d)f(x) = cf'(x)$. If $c \neq 0$, then $f = ce^{\lambda x}$, and the equation forces $c\lambda = \frac{1}{2}$: $f(x) = \frac{e^{\lambda x}}{2\lambda}$, $\lambda \neq 0$.

Hint 2. If $c = 0$, either $f \equiv 0$ or $d = 1$. Regularity: if $f(y_0) \neq 0$, then $f'(x) = \frac{f(x+y_0)-f(x)f'(y_0)}{f(y_0)}$ is differentiable, so $f \in C^\infty$. Differentiate in y at $y = 0$: once gives $f''(0) = 0$, twice gives $f'' = kf$ with $k = f'''(0)$.

Hint 3. With $f(0) = 0$, $f'(0) = 1$: $f(x) = x$, $\frac{\sin \omega x}{\omega}$ or $\frac{\sinh \omega x}{\omega}$ ($\omega > 0$), and all of them work. Answer: $f \equiv 0$, $\frac{e^{\lambda x}}{2\lambda}$ ($\lambda \neq 0$), x, $\frac{\sin \omega x}{\omega}$, $\frac{\sinh \omega x}{\omega}$.

Problem 4

Hint 1. Let $g = f^2 \geqslant 0$. Then $g' = f$ and $(g')^2 = g$. On an interval where $g > 0$, the derivative $f = g'$ does not vanish and, having the Darboux property, keeps its sign; so $(\sqrt{g})' = \pm\frac{1}{2}$, i.e. $\sqrt{g}$ is linear with slope $\pm\frac{1}{2}$.

Hint 2. Each component of $\{g > 0\}$ is therefore a half-line: a linear function vanishes at most once, so the component cannot have two finite ends. On (b, ∞) we get $f(x) = \frac{x-b}{2}$, on $(-\infty, a)$ we get $f(x) = \frac{x-a}{2}$.

Hint 3. Answer: for $-\infty \leqslant a \leqslant b \leqslant +\infty$, $f(x) = \frac{x-a}{2}$ for $x < a$, $f = 0$ on $[a, b]$, $f(x) = \frac{x-b}{2}$ for $x > b$ (all of these work; $a = -\infty$, $b = +\infty$ gives $f \equiv 0$).

Problem 5

Hint 1. For $x > 1$, $f'(x) = f(x) - f(x-1)$. Inductively, if $f \in C^{k-1}$ on $(k-1, \infty)$, the right-hand side is C^{k-1} on (k, ∞), so $f \in C^k$ there. More cannot be said: f on $[0, 1]$ is an arbitrary continuous function with $f(1) = \int_0^1 f$.

Hint 2. Substituting $e^{\lambda x}$ into the integral equation gives $\int_0^1 e^{-\lambda s}\,ds = 1$, i.e. $\lambda = 1 - e^{-\lambda}$. The root $\lambda = 0$ gives the constants. If $\operatorname{Re}\lambda \geqslant 0$ and $\lambda \neq 0$, then $\left|\int_0^1 e^{-\lambda s} ds\right| < 1$; so all other roots have negative real part, and the corresponding modes decay.

Hint 3. Every solution is bounded: $f(x)$ is the average of f over $[x-1, x]$, so $\sup_{[0,x]} |f| = \max_{[0,1]} |f|$. Moreover $I(x) = \int_{x-1}^{x} (t - x + 1) f(t)\,dt$ is constant ($I' = f(x) - \int_{x-1}^{x} f = 0$). By the bounded variation from Definite integrals – 4, f has a limit L, and $I \equiv \frac{L}{2}$ gives $L = 2\int_0^1 t f(t)\,dt$.